Covariant-Complete Dual Tetrad Gravity: A Zero-Parameter Lagrangian from Topological Field Closure

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Supplemental Material to "Dual Tetrad Gravity And The Field Origin"

Summary. The dual tetrad topology derived in the manuscript [6] forces a finite coupling channel count, which imposes a saturation condition on the gravitational field. Saturation, combined with detailed balance, uniquely determines the constitutive law $\mu(x) = x/(1+x)$, which integrates to a unique scalar Lagrangian through the standard AQUAL variational framework [5]. This supplemental note presents the resulting covariant-complete, zero-parameter scalar-tensor action, its derivation from the topology, its field equations, and its predictions.

I. From Topology to Action

The manuscript [6] establishes that ghost-free bimetric gravity [4], formulated in vielbein variables, has a coupling topology given by two tetrahedra with opposite orientations sharing a common center. The Hassan–Rosen frame transition matrix $\mathcal{S}^a_b$ is identically the coupled face matrix of this dual tetrad topology, and parity, metric exchange symmetry, and field closure determine all five β_n coupling constants with zero free parameters. The topology has $k = 4$ faces per tetrahedron and $k^2 = 16$ independent coupling channels.

From this topology the gravitational constant follows [6]:

$$G = \frac{17}{13}\,\alpha^{21}\,\frac{\hbar c}{m_e^2} \tag{1}$$

The exponent $21 = 1+k+k^2$ counts the coupling depth; the prefactor $17/13$ is the ratio of coupling channels to structural elements. The predicted value agrees with measurement to 0.12% at tree level; the Schwinger one-loop correction $\alpha/(2\pi)$ reduces the residual to 0.001% [6]. Every integer is counted from the stellated octahedron; none is adjustable.

The derivation chain. The dual tetrad topology *is* field closure: the requirement that a distributed gravitational field settle to a consistent configuration at every point [1]. The finite channel count ($k^2 = 16$) imposes a saturation condition on gravitational coupling: at weak field gradients, the field cannot recruit channels beyond those available, and the response flattens. Saturation, combined with detailed balance, determines a unique constitutive law. The constitutive law then integrates to a unique Lagrangian via the AQUAL variational relationship [5]. The result is a covariant action with disformal electromagnetic coupling in which every component is derived from the topology. The action contains zero free parameters.

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}\,\mathcal{F}\!\left(\frac{|\nabla\Phi|^2}{a_0^2}\right) - \frac{1}{4}\,e^{-2\Phi/c^2}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{\text{matter}}\right] \tag{2}$$

Each term has a definite origin:

- $R/(16\pi G)$: spacetime curvature (Einstein–Hilbert), with G given by Eq. 1
- $\mathcal{F}(y)$: gravitational saturation from field closure (derived below)
- $e^{-2\Phi/c^2} F_{\mu\nu}F^{\mu\nu}$: disformal electromagnetic coupling; photons propagate on the effective metric $\tilde{g}_{\mu\nu} = e^{-2\Phi/c^2}\, g_{\mu\nu}$, recovering the full GR deflection angle $\theta = 4GM/(rc^2)$ in the weak-field limit
- $\mathcal{L}_{\text{matter}}$: baryonic sources

From saturation to Lagrangian. At strong field gradients (well above the topological scale a_0), all coupling channels are fully active and the field responds linearly; this is the Newtonian regime. At weak field gradients (well below a_0), the finite number of coupling channels saturates: the field cannot recruit additional channels, and the gravitational response flattens. At equilibrium, two competing rates govern the partition between resolved and unresolved coupling capacity. A fraction ε of the available channels successfully tracks the Newtonian demand; the remainder $(1-\varepsilon)$ has saturated. The resolution rate is $W_+ = \kappa \cdot x \cdot (1-\varepsilon)$, driven by gravitational intensity $x = |\nabla\Phi|/a_0$ but limited by available capacity. The relaxation rate is $W_- = \kappa \cdot \varepsilon$, proportional to the resolved fraction. Setting $W_+ = W_-$ (detailed balance) gives $\varepsilon = x/(1+x)$, so the effective gravitational response is [7,8]:

$$\mu(x) = \frac{x}{1+x}$$

where $x = |\nabla\Phi|/a_0$ measures the field gradient in units of the topological scale. This result is unique within the class of single-channel saturating closures with linear response [7]: the driving exponent is forced to be 1 by the deep-MOND limit, and the "1" in the denominator is the always-present baseline coupling channel (the Field Origin). This function has the correct limits: $\mu \to 1$ (Newtonian) when $x \gg 1$, and $\mu \to x$ (saturated) when $x \ll 1$.

The scalar Lagrangian $\mathcal{F}$ is then *derived* by integrating this constitutive law. The Euler–Lagrange equation for the scalar field Φ relates the Lagrangian to the constitutive law through the condition $d\mathcal{F}/dx = x\,\mu(x)$ (the standard AQUAL variational relationship [5]), where $y = x^2 = |\nabla\Phi|^2/a_0^2$. Integrating:

$$\frac{d\mathcal{F}}{dx} = \frac{x^2}{1+x} \quad \longrightarrow \quad \mathcal{F}(y) = \tfrac{1}{2}y - \sqrt{y} + \ln(1+\sqrt{y}) \tag{3}$$

This integration is unique: the integration constant is fixed by $\mathcal{F}(0) = 0$, and the constitutive law admits no alternative form under the field closure boundary condition. The Lagrangian is the unique output of a requirement that the topology imposes. The full derivation of $\mu(x)$ from closure constraints appears in [7]; the covariant embedding in a scalar-tensor framework is developed in [8]; and the manuscript [6], Section VII.A, presents the complete treatment.

Limit checks. In the Newtonian regime ($y \gg 1$): $\mathcal{F}(y) \to \frac{1}{2}y$, recovering the standard kinetic term and reproducing GR. In the deep-MOND regime ($y \ll 1$): $\mathcal{F}(y) \to \sqrt{y}$, producing the modified dynamics observed in galactic rotation curves [3]. The logarithmic term encodes the smooth crossover between regimes. The action reproduces GR where GR is well-tested and produces MOND phenomenology where dark matter is otherwise required, with correct asymptotic behavior in both limits.

Field equations. Varying the action (Eq. 2) with respect to the scalar field and the metric produces three sets of field equations:

1. A *modified Poisson equation* governing galactic-scale gravitational dynamics

2. *Modified Friedmann equations* governing cosmological expansion

3. A *growth equation* governing cosmological structure formation

These field equations predict the CMB gravitational enhancement factor (the ratio of total gravitating mass to baryonic mass at recombination) as $1/\mu(x_{\rm rec}) = 6.46$, matching Planck 2018's measured value of 6.40 to 1%, with zero free parameters.

There are zero free parameters in this action. The Lagrangian $\mathcal{F}$ (Eq. 3), the acceleration scale $a_0 = k^2 G m_e / r_e^2$, and the gravitational coupling G are all determined by the dual tetrad structure. Unlike ΛCDM, which requires six fitted cosmological parameters and a dark matter component, this action is fully specified by the topology.

II. Zero-Parameter Predictions

The action and its field equations produce the following predictions:

Quantity	**Predicted**	**Measured**	**Deviation**
G (gravitational constant)	6.666×10^{-11} $\mathrm{m^3 kg^{-1} s^{-2}}$	6.674×10^{-11}	0.12%*
H (Hubble constant)	69.92 km/s/Mpc	70.0 ± 12	0.11%
m_e (electron mass)	9.11×10^{-31} kg	9.109×10^{-31}	0.10%
a_0 (MOND scale)	1.20×10^{-10} $\mathrm{m/s^2}$	1.20×10^{-10}	0.41%
CMB enhancement	6.46	6.40 (Planck 2018)	1.0%

*Tree-level deviation; the Schwinger one-loop correction $\alpha/(2\pi)$ reduces this to 0.001% (see [6]).

Five predictions across five domains, all with zero adjustable parameters. Changing the coupling exponent from 21 to 20 or 22 shifts the predicted G by a factor of 137 ($= 1/\alpha$); only the counted value produces agreement.

III. Acknowledged Limitations

The framework does not address:

- Galaxy cluster dynamics (the Bullet Cluster and residual cluster mass)
- The full CMB angular power spectrum shape (the enhancement factor is predicted to 1%, but the detailed multipole structure has not been computed)
- The strong and weak nuclear forces
- The cosmological constant

The tree-level 0.12% residual in G is identified in [6] as the Schwinger one-loop correction $\alpha/(2\pi)$; including it reduces the residual to 0.001%. The manuscript [6] also demonstrates that the dual

tetrad topology constrains the coupling parameters of the established Hassan–Rosen bimetric action [4]; the relationship between that bimetric formulation and the scalar-tensor action presented here is identified as a target for future work.

References

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